

NATIONAL UNIVERSITY OF COMPUTER AND EMERGING SCIENCES

Applied HCI

Deliverable 3: Alternatives, Selection Rationale and Storyboard

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1. Purpose, inputs from D2 and AI use

D3 records how we moved from "what could we build" to "what will we build, and why". We first listed many different ways of solving the D2 problem, then critiqued every one of them against the D2 requirements, and only then chose a direction. Every decision below is traced to a D2 requirement, theme or evidence note (IDs such as H-15 refer to the D2 sticky notes, which in turn refer to D1B).

Core task (from D2): setting up and starting an unfamiliar household appliance for its first task, using its printed manual. Observed in D1B on a Panasonic top-load washing machine (F, H, G) and a mechanical-dial air fryer (I, J).

1.1 What we carried forward from D2

D2 item	What it says	How D3 uses it
POV-1 / HMW-1 (primary)	Persona A needs to match every named part and symbol to the real machine from where they stand, before acting (T1).	Primary problem for ideation and the main selection criterion.
HMW-2	Tell a normal machine response from a fault at the moment it happens (T2).	Secondary question.
HMW-3	Make the instructions the quickest place to confirm a step, so users do not leave for a video or relative (T3, T5).	Secondary question.
HMW-4	Turn the stated limits (13.5 kg, MAX line, 10 cm, preheat 3 to 5 min) into checks users can do by eye (T4).	Secondary question.
Personas	A: Careful First-Timer (H, I), reads the manual from page one. B: Lookup User (F, G), machine first then one table or number. C: Try-First Fixer (J), tries the machine, then YouTube.	Every concept is checked for which persona it actually reaches.
R1	Identify each named part, control and symbol correctly first time, without help (4 of 5; 0 wrong-compartment pours; 0 basket removals without lifting the guard).	Critique and scoring criterion.
R2	Reach the instruction for the current step in 60 s or less, within 3 page turns, for both page-one and lookup readers.	Critique and scoring criterion.
R3	Correct order of steps and no cautioned action (no release while shaking, no overfill, preheat before loading, detergent in main-wash before Start).	Critique and scoring criterion.
R4	Understand normal responses when they happen, finish alone, feel confident (0 fault readings; 1 of 5 or fewer seek outside help; confidence 6 or more of 7).	Critique and scoring criterion.
R5	Critical markings perceivable and understandable regardless of print size, eyesight and language (symbols paired with words; contrast 4.5:1 or better; English reachable without paging).	Critique and scoring criterion.

1.2 AI use declaration

Component	AI stance in the brief	What we did
1. Individual alternatives	Allowed	All 20 concepts in Section 2 (five per member) were generated by the team members themselves, using the D2 report as input. AI (Claude, Anthropic) was used only to help format and structure the document and to reword our ideas for clarity. No concepts were created by AI, and each set was reviewed and edited by the member named on it.

Component	AI stance in the brief	What we did
2. Team convergence record	Allowed	The critiques and the scoring grid were written by the team. AI was used only to help format and structure the grid and to reword our critique wording. Every score is justified against the D2 baselines and sticky-note evidence, and each concept is critiqued the same way, whether or not AI helped to reword it.
3. Storyboard	Allowed	Rough panels were built by the team from plain shapes and text in a draw.io style, so they can be redrawn on paper. No generated artwork was used, and AI was not used for the storyboard content.
4. Design rationale	Allowed	The reasons for selection are traced to D2 requirements and D1B evidence notes, not to preference. The reasoning is our own; AI was used only to help structure the section and reword our ideas.

2. Individually generated alternatives

Each member contributed five concepts, generated individually before any team discussion. The ideas are not grouped by theme: every member's set mixes printed, on-machine, machine-behaviour and phone-based ideas, and some ideas overlap on purpose (for example, M1-A and M2-C both match a part on the page to a mark on the machine). The last column of each row is the one-line link to the D2 requirement(s) the concept addresses. Concept IDs: M1 = Muneeba, M2 = Armish, M3 = Areej, M4 = Warisha.

ID	Concept	D2 requirement(s) addressed	Member
M1-A	Colour pairs. Each critical part gets a bright colour on the machine, and the same colour appears in the manual drawing and step.	R1 (match at a glance), R3 (cautioned parts stand out).	Muneeba Zaheer
M1-B	"Is this normal?" fold-out. A page listing every sound, display value and pause in the first 10 minutes (lid click, "32", drawer pause, silent heating), each marked normal or stop.	R4 (normal response understood).	Muneeba Zaheer
M1-C	Plain display. At power-on the display scrolls "WASH TIME 32 MIN" instead of the bare number.	R4 (the "32" is understood), R5 (words).	Muneeba Zaheer
M1-D	First-Use Guide (QR, phone). A QR on the machine and on manual page one opens a model-specific guide, one step per screen: a photo with the part ringed, the action, a "you should now see or hear" line, and each limit shown as a picture.	R1, R2, R3, R4, R5.	Muneeba Zaheer
M1-E	Hand-and-eye guide. A life-size printed guide that turns limits into checks: hand span against the 10 cm clearance, a drum picture of "full to here", an "empty first" preheat pictogram.	R3 (no overfill, preheat order), R5 (visual, no tools).	Muneeba Zaheer
M2-A	First-Use Card. One sheet in English only, titled for the task ("First wash"), with six numbered steps in the order the machine needs them, kept in the box.	R2 (steps reached at once), R3 (order shown), R5 (English first).	Armish Tauqir
M2-B	Status light and voice. After each state change, a light and a short phrase in English or Urdu: "Lid locked. Normal." / "Drawer open. Paused. Normal."	R4 (normal response at the moment), R5 (user's language).	Armish Tauqir
M2-C	Matching number stickers. Numbered stickers on each critical part (tray compartments, guard, release, MAX line) using the same numbers as the manual drawing.	R1 (same number on page and part), R5 (words with numbers).	Armish Tauqir
M2-D	Delivery demo. An installer runs a five-minute first-use walkthrough at delivery and leaves a checklist.	R1, R3, R4 (all partly, shown on the real machine).	Armish Tauqir
M2-E	Moulded fill band. A wide, high-contrast fill band built into the drum and basket, with a clear STOP mark readable at a glance.	R3 (no overfill), R5 (visible mark), R1 (MAX line seen).	Armish Tauqir

ID	Concept	D2 requirement(s) addressed	Member
M3-A	Standing-angle photo pages. Replace side-view line drawings with photos taken from where the user stands, each critical part ringed and named with the same word printed on the machine.	R1 (match part to photo), R5 (larger, labelled).	Areej Fahim
M3-B	Model-specific 60-second video. A QR on the machine opens one short silent captioned video for that exact model.	R2 (partly), R4 (partly), R1 (partly).	Areej Fahim
M3-C	Lid insert. A printed word-and-icon plate beside the detergent tray: MAIN WASH / SOFTENER, with a one-line "pour here first" prompt.	R1 (right compartment), R3 (detergent order), R5 (symbols with words).	Areej Fahim
M3-D	Preheat-done signal. The air fryer shows "Preheating, empty", then sounds a tone and shows "Add food now".	R3 (preheat before loading), R4 (a "done" signal exists).	Areej Fahim
M3-E	Guided first-use mode. The panel lights step 1 to 6 in order and waits for each action before lighting the next.	R1 (partly), R2 (no hunting), R3 (order), R4 (confirms each response), R5 (partly).	Areej Fahim
M4-A	Two-door manual. Page one asks "start from the beginning, or jump to your step?"; the contents use task verbs; other languages go in separate booklets.	R2 (both reading habits), R5 (English reachable).	Warisha Shaukat
M4-B	Load check. The drum weighs the load and displays "About half / Full / Too much".	R3 (no overfill), R4 (confidence about load).	Warisha Shaukat
M4-C	Numbered pull-tabs. Large tabs on the air-fryer guard and release ("1 lift, 2 press") so the order is built into the part.	R1 (guard recognised), R3 (guard before release), R5 (labelled).	Warisha Shaukat
M4-D	Ask with a photo. The user sends a photo of the panel or part to a brand helper and gets a plain-language answer.	R4 (response explained), R2 (partly), R1 (partly).	Warisha Shaukat
M4-E	Camera overlay. The phone camera names each part on the live machine image and rings the part the step refers to.	R1 (match from the user's own angle), R5 (labels).	Warisha Shaukat

How distinct are the 20? They differ in where the help lives: on the printed page (M1-B, M1-E, M2-A, M3-A, M4-A), on the machine body (M1-A, M2-C, M2-E, M3-C, M4-C), in the machine's own behaviour (M1-C, M2-B, M3-D, M3-E, M4-B), on the phone (M1-D, M3-B, M4-E), or with a person (M2-D, M4-D). Where concepts share a mechanism we say so: M1-A and M2-C are two forms of one on-machine matching idea; M1-D and M3-B both start from a QR but differ in structure (step by step versus watch); M1-D and M4-E both use the phone, but M1-D shows a fixed photo while M4-E works on the live camera view.

3. Team convergence record

3.1 Method

We critiqued all 20 concepts in the same way, whatever their author or our first reaction. For each one we wrote one strength, one concern, and the single requirement it satisfies best. Strengths and concerns point to D2 evidence notes. Then we scored each concept against R1 to R5 (Section 3.3) before any choice was made. The table is ordered by concept ID, and ends with the two concepts that led the field, M3-E and M4-E.

3.2 Structured critique of every alternative

ID	Strength (with evidence)	Concern (with evidence)	Best satisfies
M1-A	A fast visual match at a glance (F-08).	Colour alone fails readers with colour-vision difficulty and prints badly in black-and-white booklets (R5).	R1
M1-B	I-11 shows that a forewarning stops the "Did I break it?" reaction.	The answer must be looked up at the moment of surprise, and H-14 and J-06 reacted in seconds.	R4

ID	Strength (with evidence)	Concern (with evidence)	Best satisfies
M1-C	Removes "32 what? Minutes?" (F-03, H-08).	Needs a firmware change, and the air fryer has no display.	R4
M1-D	One step-by-step flow touches all five requirements and reaches the try-first user, who opened YouTube before the manual (J-04).	The photo is taken from one fixed angle, while I-05 shows the mismatch is about the user's own viewing angle, so R1 is only partly met. Needs a phone, signal and noticing the QR; QR-linked guides are already common. Cannot enlarge the tiny symbols on the machine (H-13).	R2 (also R4)
M1-E	Tackles numbers users cannot measure (H-05, I-06).	A printed guide only approximates, and a cautious user may still under-load (H-06).	R3
M2-A	Takes page-one and lookup readers straight to ordered steps without warnings (H-15, F-12).	Persona C throws booklets away (J-01), so a loose card never reaches him.	R2
M2-B	Says "normal" at the instant of the lid click or drawer pause (H-14, J-06).	New hardware; a speaker in a noisy laundry room; repeated beeps may annoy.	R4
M2-C	Works even if the manual is never opened, so it helps Persona C (J-10); one number on page and part removes the guessing (F-02, G-06).	Stickers peel or get removed, and they cover parts only, not machine responses.	R1
M2-D	Shows parts and responses live on the real machine.	I and J's households already owned the unit, so no delivery demo would reach them; one-off and costly.	R1
M2-E	Makes the MAX line visible (I-07) and capacity a visual judgement (F-04, H-06).	Means changing moulded parts, which we cannot build or test; fixes loading only.	R3
M3-A	Answers the side-view versus top-view mismatch directly (I-05, I-03).	Only reached by people already reading the manual; tiny symbols on the machine stay tiny (H-13).	R1
M3-B	Matches what F and J actually did (F-10, J-04: video, captions, no sound).	Watched in a block, not at the step; F-09 shows a video of a different model; one camera angle.	R2 (partly)
M3-C	Fixes the exact place where three washing-machine users struggled (G-07, H-11, H-12).	Covers one tray on one machine; the air-fryer breakdowns are untouched.	R1
M3-D	Fills the missing "preheat done" signal (I-10) and the empty-or-loaded gap (I-09).	Air fryer only; J had already skipped preheat before any signal existed (J-05).	R3
M4-A	One booklet serves both reading habits (H-15, F-12).	Gets to the step faster but does nothing to match the part or judge the limit once there.	R2
M4-B	Ends "How many kilos is this?" (H-05).	Weight sensing is costly hardware and applies to the washing machine only.	R3
M4-C	Builds the guard-then-release order into the part (J-07, J-10).	Air fryer only, and it is a hardware change that tabs may snag or tear off.	R3
M4-D	Mirrors what H did with her sister (H-09), but with an answer.	It is still outside help (R4 allows 1 of 5 or fewer), and H-09 shows waiting for a reply.	R4
M3-E	The widest single fit on the machine itself: gives the order and confirms each response where it happens.	Needs a full redesign of the control panel that we cannot build or test; cannot help units already in households (I, J).	R4
M4-E	Names parts on the live view from the user's exact angle, the core of T1 (I-05, I-03, G-07); once pointed at a control it can also say "normal" at that moment (H-14, J-06). Reaches the try-first user, who picks up a phone before the manual (J-04).	Needs per-model recognition that is hard to prototype at low fidelity; hands may be full; needs a phone; the tiny markings on the machine stay tiny (H-13); Persona A may stay on paper (I used no phone).	R1 (also R4)

3.3 Scoring against R1 to R5, persona reach and scope gate

Scoring rule. For each requirement: 2 = directly addresses the D1B breakdowns behind that requirement's success criterion; 1 = addresses part of it (one appliance, one breakdown, or only if the user chooses to engage); 0 = does not address it. Maximum 10.

Persona reach. Which of the three personas would actually meet the concept on the path D2 observed: A (manual first), B (machine first, then one lookup), C (machine first, then YouTube).

Scope gate (pass or fail). Can we build a rough version and test it in D5A/D5B on the same two appliances without changing the appliance's own hardware or firmware? This is a course constraint (D4A wireframes and D5 testing), not a view on which idea is better. A concept that fails the gate can still be a good idea for a manufacturer.

ID	Concept	R1	R2	R3	R4	R5	Total	Personas reached	Scope gate
M1-A	Colour pairs	2	0	1	0	0	3	A, B, C	Pass
M1-B	"Is this normal?" fold-out	0	1	0	1	0	2	A, B	Pass
M1-C	Plain display	0	0	0	1	1	2	A, B	Fail (firmware)
M1-D	First-Use Guide (QR)	1	2	1	2	1	7	A, B, C	Pass
M1-E	Hand-and-eye guide	0	0	1	0	1	2	A, B	Pass
M2-A	First-Use Card	1	2	1	1	1	6	A, B	Pass
M2-B	Status light and voice	0	0	0	2	1	3	A, B, C	Fail (hardware)
M2-C	Number stickers	2	0	1	0	1	4	A, B, C	Pass
M2-D	Delivery demo	1	0	1	1	0	3	A, B (new buyers only)	Pass
M2-E	Moulded fill band	1	0	2	0	1	4	A, B, C	Fail (moulded part)
M3-A	Standing-angle photo pages	2	0	1	0	1	4	A, B	Pass
M3-B	60-second video	1	1	1	1	1	5	A, B, C	Pass
M3-C	Lid insert	1	0	1	0	1	3	A, B, C	Pass
M3-D	Preheat-done signal	0	0	2	1	0	3	A, C	Fail (hardware)
M4-A	Two-door manual	0	2	0	0	1	3	A, B	Pass
M4-B	Load check	0	0	2	1	0	3	A, B	Fail (hardware)
M4-C	Numbered pull-tabs	1	0	1	0	1	3	A, C	Fail (hardware)
M4-D	Ask with a photo	1	1	0	1	1	4	A, B, C	Pass
M3-E	Guided first-use mode	1	2	2	2	1	8	A, B, C	Fail (control panel)
M4-E	Camera overlay	2	1	1	2	1	7	A, B, C	Pass (hard to prototype)

Scoring notes. (1) M4-E is scored as refined after critique (Section 3.5). As first written (name each part and ring the part the step refers to, nothing more) it would have scored R1 2, R2 0, R3 0, R4 0, R5 1 = 3. (2) M1-D scores 1 on R1, not 2: its photo is shot from one fixed angle, and I-05 shows the breakdown is the mismatch with the user's own viewing angle, so a fixed photo answers it only in part. (3) M4-E scores 1 on R2 because the user has to open the phone and choose the task, and R2 is untimed in D1B.

3.4 What the record shows

Ranked by total score, **M3-E leads with 8 of 10**, but it fails the scope gate (it needs a redesigned control panel). Among the concepts that pass the gate, **M4-E and M1-D tie at 7**, followed by M2-A (6) and M3-B (5). Because the

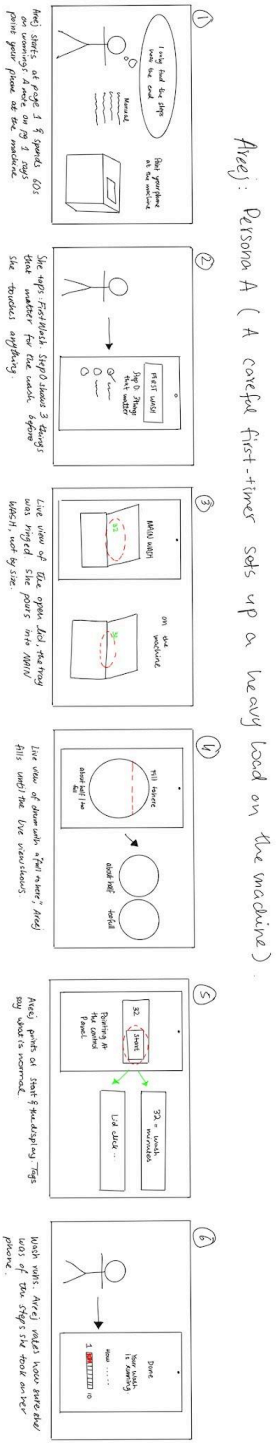
tie cannot be settled on the total, we settled it on the primary problem: D2 names POV-1 / HMW-1 (T1, recognising the part from the user's own angle) as the primary problem, and T1 is the strongest theme (15 notes, 5 of 5 participants). M4-E scores 2 on R1 and M1-D scores 1. M1-D scores 2 on R2, but D1B never timed R2 for every participant, so that advantage is less well evidenced than the R1 gap. Section 5 gives the full reasoning.

3.5 Refinement of M4-E after critique

The critique showed that a camera overlay that only names and rings parts is strong on R1 but silent on R2 to R4. We therefore refined it before scoring, borrowing the structure that worked in M1-D and M3-E: it opens on a task choice ("First wash") and shows one step at a time, with the cautions for that step; it tags normal responses when the user points the camera at a control ("32 = wash minutes. Normal." / "Lid click = lid locked. Normal."); and it draws a "full to here" line on the drum. The part-naming on the live view is unchanged.

4. Storyboard

The storyboard follows **Persona A, the Careful First-Timer** (grounded in H and I), the persona of POV-1, through the core scenario "set up and start a wash for a heavy load" on the washing machine. It uses the selected concept, **M4-E (camera overlay)**. The six panels cover the trigger, three key interaction moments and the resolution. They are rough structure-and-flow sketches built from boxes, stick figures and phone outlines. Quotes and behaviours come from the D2 notes.



5. Design rationale

5.1 Concept summary

The **camera overlay** is a phone view of the real machine. It names each part in the machine's own words and rings the part the current step refers to, from the angle the user is standing at. It opens on the task ("First wash"), shows one step at a time with that step's cautions, tags normal responses ("32 = wash minutes. Normal.") when the user points at a control, and draws a "full to here" line on the drum.

5.2 Requirements addressed, and how

Req.	How the camera overlay addresses it	Score	What it leaves open
R1	The live view names each critical part (tray compartments, guard, release, MAX line) in the machine's own word and rings the one the step names, from the user's own position. This answers the side-view versus top-view mismatch and the hidden or unlabelled parts (I-05, I-03, G-07, H-11, J-10).	2	Needs per-model recognition. Cannot force the user to open the phone before touching the machine.
R2	Opens on a task choice and shows one step per screen, so the current step is one or two taps away whether the user starts at page one or at the machine (H-15, F-12).	1	The user must open the phone and pick the task. The 60 s target must be timed in D5B; D1B did not time it.
R3	Steps come in the right order (air fryer: preheat before loading) and each step carries its cautions (no release while shaking, J-07). A "full to here" line on the drum replaces a kilo figure (H-05, F-04).	1	Cannot stop an action taken before the overlay is opened.
R4	When the user points at Start, the display, the lid or the drawer, a tag says what it is and that it is normal ("32 = wash minutes", "lid click"), at the moment it happens (H-14, J-06, H-08, F-03). The answer is in the same place, so there is less reason to leave for YouTube or family (F-07, H-09).	2	Confidence and the outside-help count need D5B measurement.
R5	Part names shown in large text with every symbol paired with words; enlarged close-ups of the tiny symbols (H-13, I-07); English first, optional Urdu.	1	The markings printed on the machine stay as small as before.

Of the HMW questions, HMW-1 (recognise the part from the user's own angle) and HMW-2 (normal versus fault) are answered directly. HMW-3 (make the instructions the quickest place to confirm a step) is answered in part through the task-first entry, and HMW-4 (limits as checks) only for the pictured limits.

5.3 Alternatives seriously considered and not selected

Alternative	What it offered	Why it was not selected
M3-E Guided first-use mode	Highest score (8). It gives the order and confirms every response on the machine itself.	Fails the scope gate: it needs a redesigned control panel that we cannot build or test in D4A to D5B. It also cannot reach units already in households, as with I and J (D2 section 1 table).
M1-D First-Use Guide (QR, phone)	Ties for the top score among concepts that pass the gate (7). Scores 2 on R2 against 1 for the overlay, and reaches all three personas.	Scores 1 on R1 against 2 for the overlay: its photo is taken from one fixed angle, which only partly answers the mismatch in I-05. R1 is the primary problem (POV-1, T1) and had the worst D1B baseline. QR-linked guides are also already common, so it adds little beyond what users can find now; we did not select on novelty. We carry its task-first, one-step-per-screen structure into the overlay.
M2-A First-Use Card	Best printed option (6). Clearly takes Persona A and B to ordered steps (H-15, F-12).	Scores 1 on R1 and R4 against 2 for the overlay; it does not reach Persona C, who throws booklets away (J-01) and tries first (J-09); and a printed card cannot show a response at the moment it happens.

Alternative	What it offered	Why it was not selected
M2-C Matching number stickers	Strong on R1 (2) and reaches all three personas, including C (J-10).	Scores 0 on R2 and R4, so it leaves two of the five requirements unaddressed. We keep it as an idea to test as an add-on in D5.

5.4 Reason for selection

We selected the camera overlay because of how it scores against D2, not because the team liked it.

1. **It answers the primary problem most directly.** POV-1 and HMW-1 ask for matching each named part to the real machine from the user's own angle (T1: 15 notes, 5 of 5 participants). In D1B the tray was not where 3 of 3 washing-machine users expected it and 0 of 2 air-fryer users recognised the guard first time (R1 baseline). A live view is the only concept that passes the scope gate and shows the part from the user's actual angle; photo pages (M3-A) and stickers (M2-C) score 2 on R1 but 0 on R4.
2. **It is the only gate-passing concept that scores 2 on both R1 and R4.** These answer the two strong themes T1 and T2 (and T3 through the shorter route to reassurance). M1-D scores 2 on R2 and R4 but only 1 on R1.
3. **It ties the best testable alternative, and the tie is broken on the evidence.** M4-E and M1-D both total 7. We chose M4-E because R1 is the primary problem with the worst baseline, whereas M1-D's R2 advantage rests on a baseline that D1B did not time for every participant.
4. **It beats the best printed alternative on the path users take.** M2-A depends on the user opening the booklet, but four of five participants did not go to the manual first (T3: H-19, G-11, J-12, F-07/F-10). The overlay meets users on the path they actually take, including J, who opened YouTube before the manual (J-04).
5. **It is testable.** M3-E scores higher (8) but fails the scope gate. M4-E can be prototyped on the same two appliances, using annotated photos and screen recordings of the real machines, with no change to the appliance.
6. **It reaches all three personas** (A, B and C), while M2-A reaches only A and B.

5.5 Open risks

Risk	Require ment	Evidence	What D4A and D5 will check
Persona A may stay on paper. Participant I used no phone during the task and her first resort was "the manual, if I can find it" (D2 section 1.3).	R2, R4	I-01; D2 section 1.3	Whether a printed fallback (an M2-A style card) is needed beside the entry mark; test with a page-one reader.
Hands may be full or wet, and a phone must be held or propped in the laundry area.	R2	D2 limitation 3 (no evidence past Start)	Large buttons, one step per screen, a stand or prop; observe use in D5B.
Live part recognition cannot be built at low fidelity, so D5 tests the concept's flow and wording, not recognition accuracy.	R1	D2 scope; T1	State this in D5 results; use annotated photos of the two D1B models; compare with the fixed-photo version (M1-D) if time allows.
Printed markings on the machine are unchanged.	R5	H-13, I-07	Enlarged close-ups of tiny symbols in the overlay; D5A accessibility inspection of text size and contrast.
The overlay cannot stop a try-first user acting early.	R3	J-02, J-07, J-09	Where the entry mark sits on the machine; whether C opens it before touching the basket.
Pictured limits (full-to line, MAX line) may not match real loads.	R3	H-05, H-06, F-04 (T4 is a supporting theme)	Check the drum picture against real loads in D5B.
Model-specific content must be correct for each machine.	R1, R4	F-09 (video of a different model)	Build and test for the two D1B models only.

Risk	Requirement	Evidence	What D4A and D5 will check
Baselines still incomplete for R2 timing and R4 confidence.	R2, R4	D2 Part 4, limitation 6	The first D5B session completes both baselines.